Priyansh Asati

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.19
Mar Thoma Gram Jyoti School , Khitola , Jabalpur, MP	Class XII (Central Board of Secondary Education)	2019	79.80%
Mar Thoma Gram Jyoti School , Khitola , Jabalpur, MP	Class X (Central Board of Secondary Education)	2017	9.60%

SKILLS

Language: C, C++, Python, SQL

ML Frameworks: Scikit-learn, PyTorch, XGBoost, Hugging Face

AI/ML: Deep Learning, Natural Language Processing, LLMs, RAG, Exploratory Data Analysis (EDA)

Data Analysis & Visualization: NumPy, Pandas, Matplotlib

Tools: Git, GitHub, Jupyter Notebook, VS Code, Google Colab, Kaggle.

Databases: MySQL

PROJECTS

Multilingual Emotion Detection | Python, PyTorch, Hugging Face Transformers, scikit-learn

- Developed a multi-label emotion classification deep learning model for English, Hindi, and Marathi text using DistilBERT and mBERT architectures.
- Increased the English model's F1-macro score from a **0.372** baseline to **0.620**, and achieved a **0.811 F1-macro** score on the multilingual dataset.
- Handled dataset class imbalance by implementing custom loss functions, specifically Focal Loss and Weighted BCE, to improve detection accuracy across all emotion categories.

CDN Simulator: Geo-Distributed Content Delivery System

- Built a CDN simulator with 3 nodes (origin + 2 edge servers) that routes requests to the nearest server based on location.
- Reduced average **response time by 8x (160ms to 20ms)** by adding caching at edge servers, tested with real request timing.
- Added automatic failover so requests reroute to a healthy server **within 1 second** if the nearest one goes down, keeping the system available during outages.
- Built **cache invalidation** with **TTL expiry** and manual purge to keep content up to date across all servers after updates.

ArXiv Paper Assistant | Python, PyTorch, Hugging Face, FAISS

- Developed the RAG system to answer questions from research paper PDFs using hybrid search (FAISS + BM25) and reranking for more accurate results.
- Built a fast search pipeline to find exact paragraphs, delivering accurate answers to user questions in just **0.22 seconds** per query.
- Achieved **94.4%** exact match accuracy by fine-tuning the retrieval and reranking models.

Hospital Resource Prediction | Python, XGBoost, LightGBM, FastAPI

- Built an automated data pipeline to extract and clean patient vitals, lab results, and demographic data from patients' first 6 hours in the ICU.
- Trained an XGBoost classification model to predict the need for mechanical ventilation, achieving an **AUROC score** of **0.725**.
- Developed a LightGBM regression model to estimate the total length of stay for patients, reaching an **R-squared score** of **0.795**

THESIS

Neuro-Symbolic VLM Architecture for Driving Risk Assessment on Unstructured Roads

- **Designing a neuro-symbolic pipeline** that grounds VLM visual reasoning with deterministic geometric scene-stats to eliminate spatial hallucinations on unstructured Indian roads.
- **Implementing Chain-of-Thought Reasoning:** Structuring a multi-step CoT prompting framework to guide the VLM through sequential hazard perception and causal reasoning, eliminating black-box predictions.

RELEVANT COURSES

Fundamentals of Machine Learning, Advanced Computer Network, Natural Language Processing, Data Science Analysis